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Department Of Computer Applications

AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies

Project Review-2

SIH ID: SIH1638

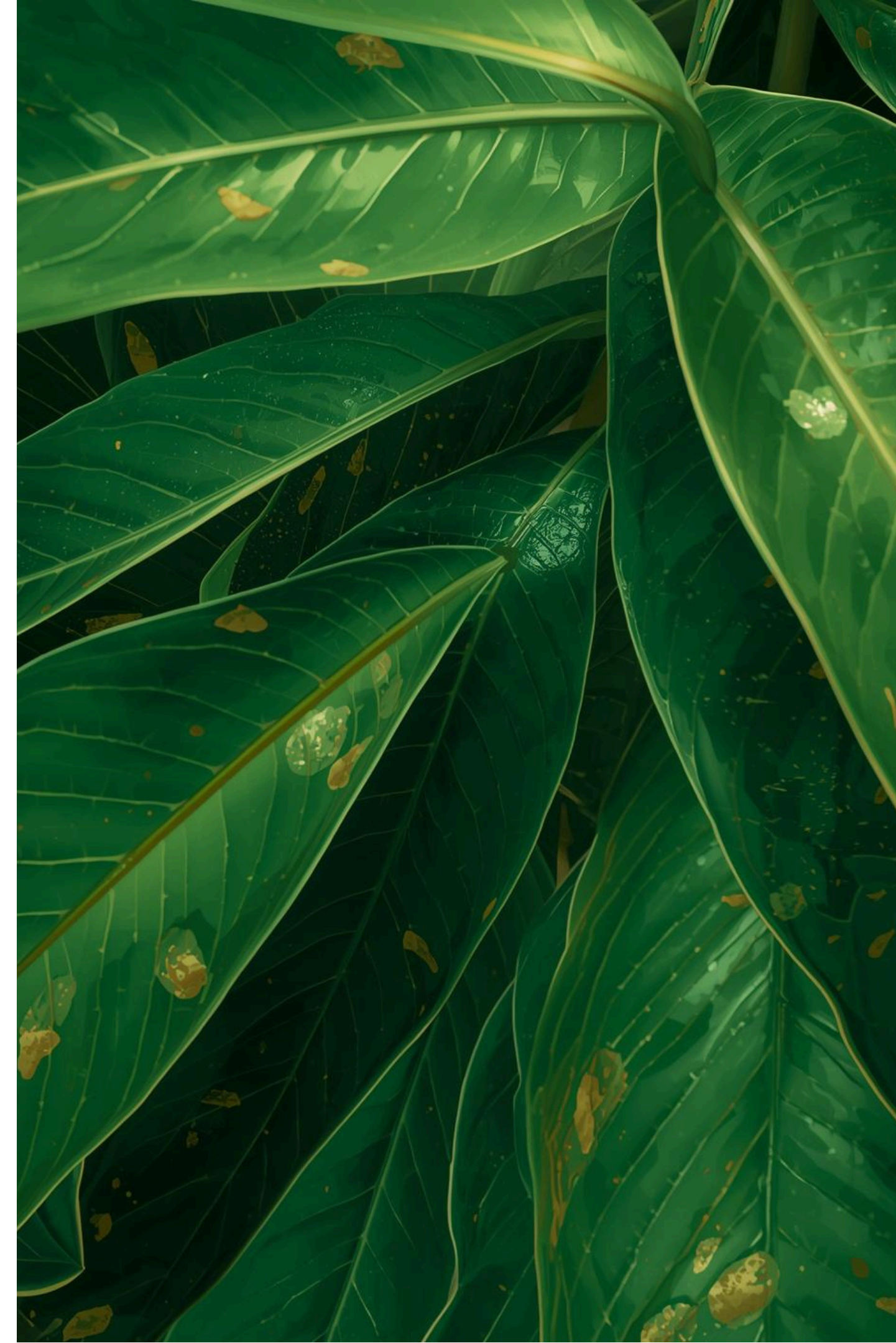
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Abstract

Mango farming often suffers yield losses due to leaf diseases and nutrient deficiencies, usually detected only after serious damage. Early and automated detection can help farmers act promptly, reduce losses, and improve crop quality. This project proposes a machine learning-based system using image processing to detect and classify diseases and nutrient deficiencies in mango leaves. Leaf images are pre-processed through resizing, normalization, noise reduction, and colour space conversion to manage variations in lighting and field conditions. Segmentation separates leaves from the background, and key features including colour histograms, texture features (GLCM and LBP), and shape descriptors are extracted. These features are analysed with machine learning algorithms such as Random Forest, SVM, and KNN for accurate classification. Later, deep learning models like DenseNet-121 with transfer learning will be explored. Unlike manual diagnosis, this approach is faster, reliable, and requires no expert knowledge, offering farmers a cost-effective and efficient tool for early detection.



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Introduction

- Mango is one of the most important fruit crops with high economic value
- Mango leaves are commonly affected by diseases and nutrient deficiencies
- Early identification is crucial to prevent yield loss and quality reduction
- Manual diagnosis is time-consuming and requires expert knowledge
- Artificial Intelligence can assist in fast and accurate leaf health analysis



Review of Literature

- We analyzed 41 research papers spanning a decade (2016–2025) on mango leaf health detection. [link](#)(literature survey excel sheet)

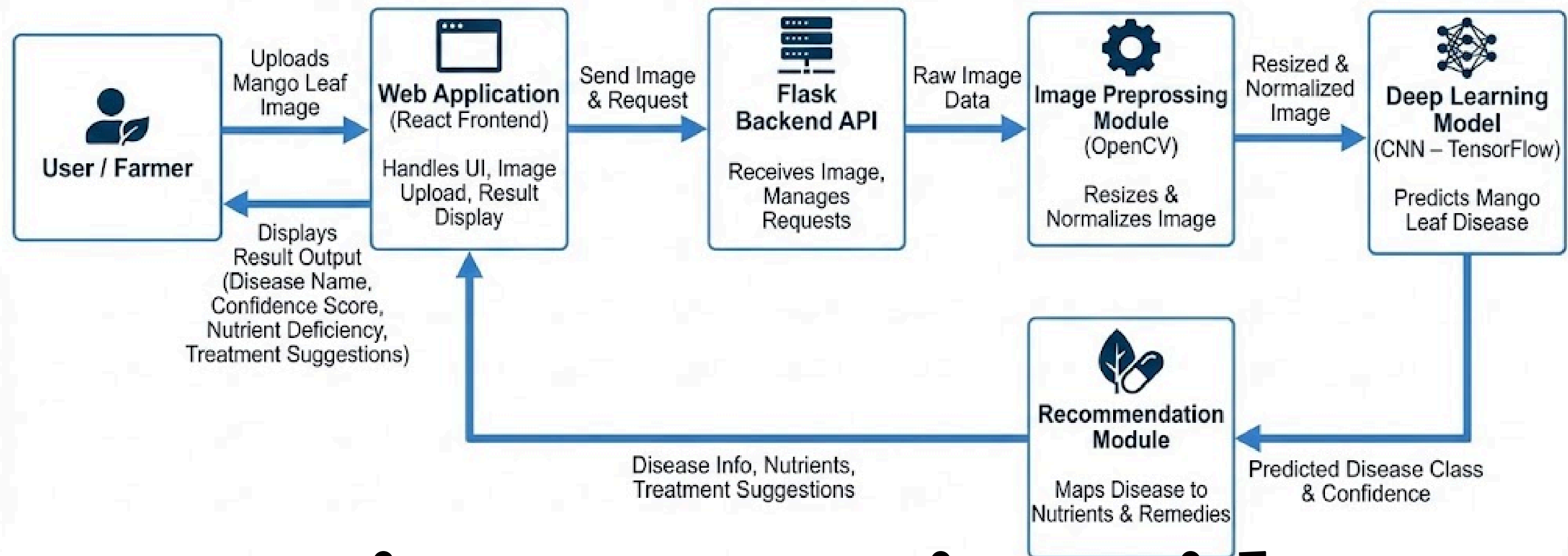
Key Findings

- **Scope Gap:** Over 90% of studies focus exclusively on disease detection, largely ignoring nutrient deficiencies.
- **Tech Evolution:** Clear shift from Classical ML (SVM, KNN) to Deep Learning (CNN, DenseNet) and now Lightweight/Explainable AI models.
- **Validation Gap:** Nearly all high-accuracy models (>95%) are validated in controlled lab conditions, not real field environments.
- **Practicality Gap:** Minimal focus on mobile deployment; only 2 papers discuss optimization for low-resource devices.
- **Our Synthesis:** This analysis identifies a critical need for an integrated, field-robust, and farmer-accessible diagnostic system.

Proposed methodology

- **Two Parallel Approaches:** Implementation of both classical machine learning and deep learning pipelines
 1. **Classical ML Pipeline:** Image preprocessing → handcrafted feature extraction → classification using SVM, Random Forest, and KNN link
 2. **Deep Learning Pipeline:** Image preprocessing → transfer learning using DenseNet-121 → multi-class classification link
- **Comparative Analysis:** Performance evaluation based on accuracy, inference speed, and computational efficiency
- **Generalization Focus:** Data augmentation and diverse dataset collection to improve robustness under field conditions

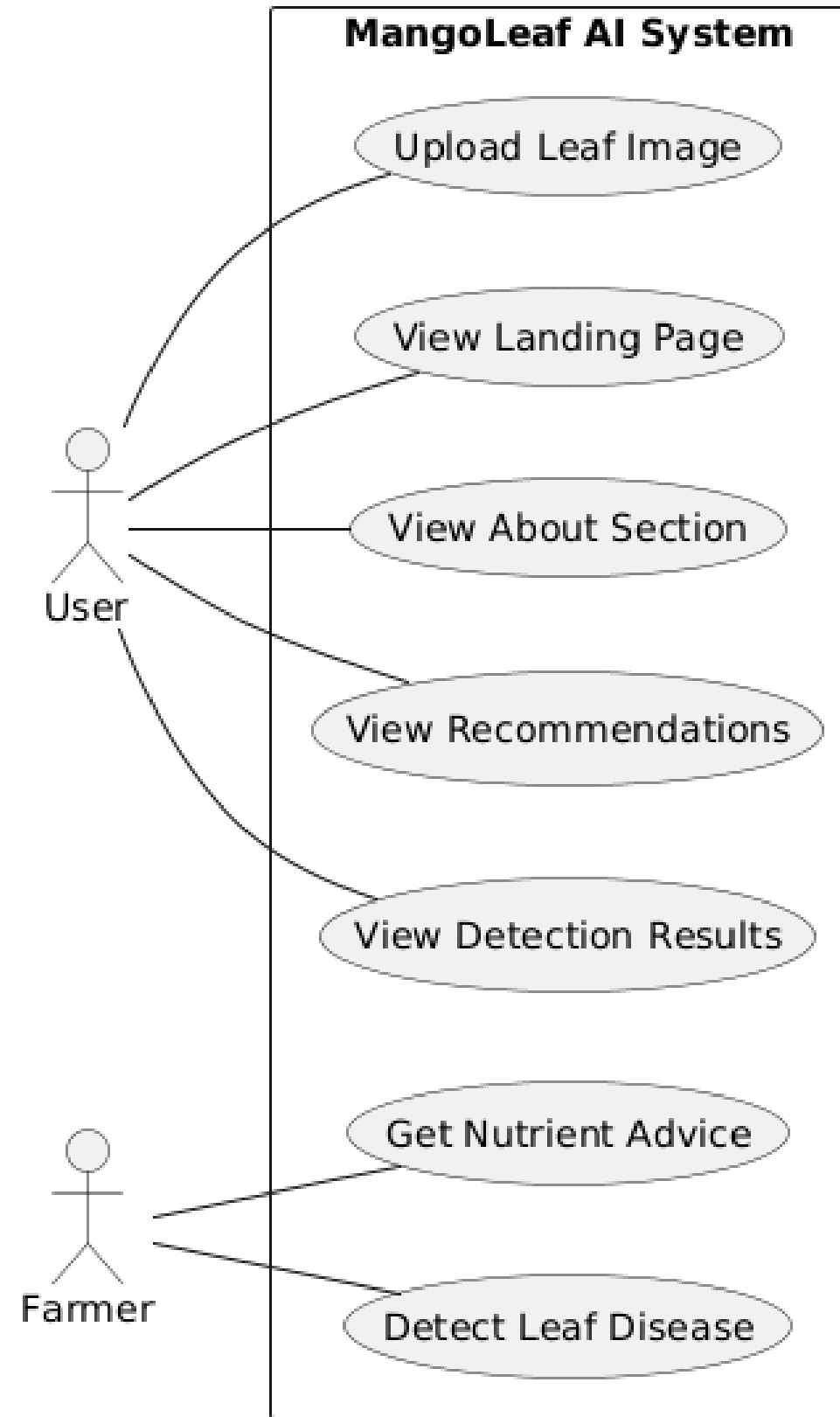
Execution



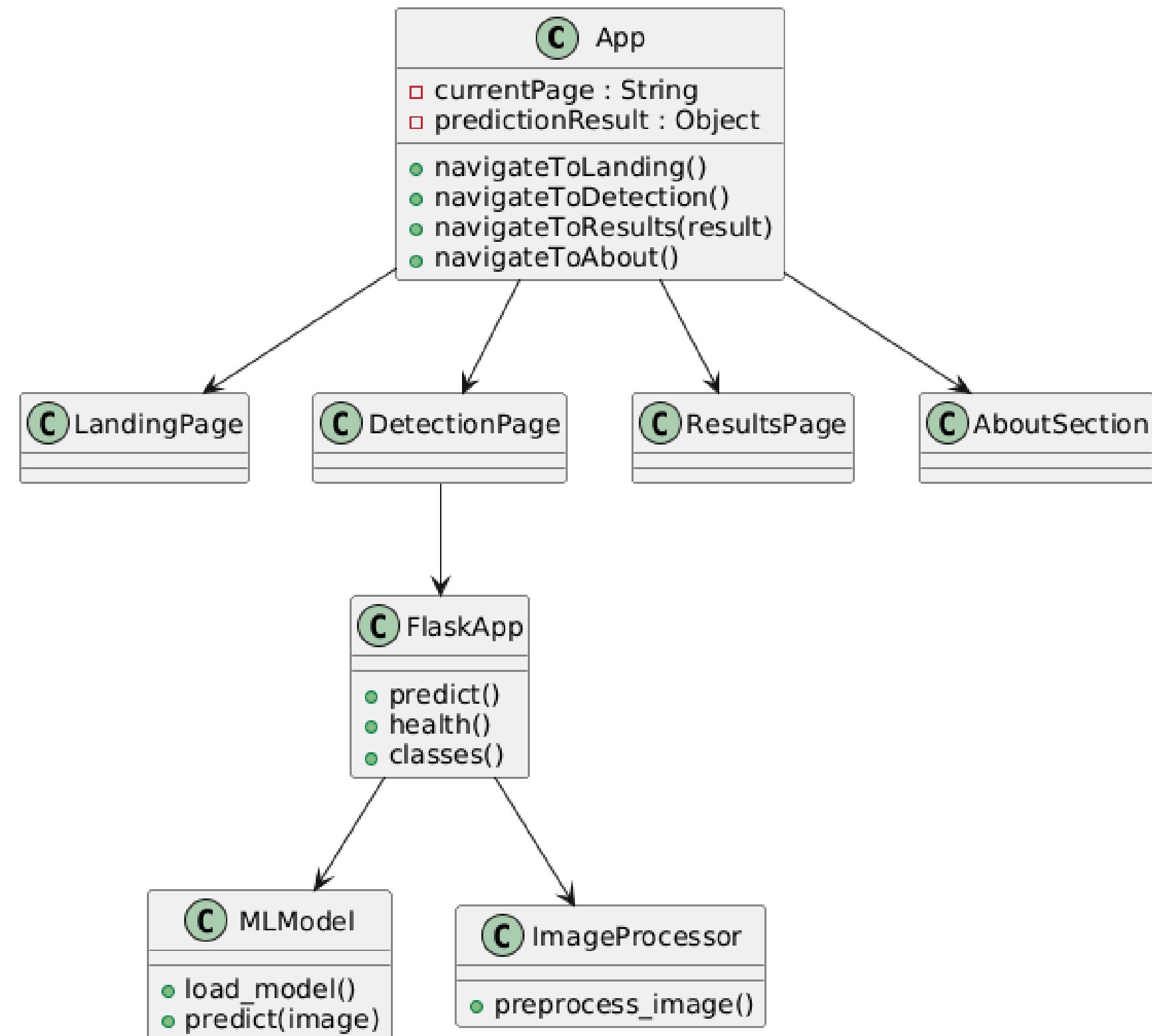
Project Execution video

UML Diagrams

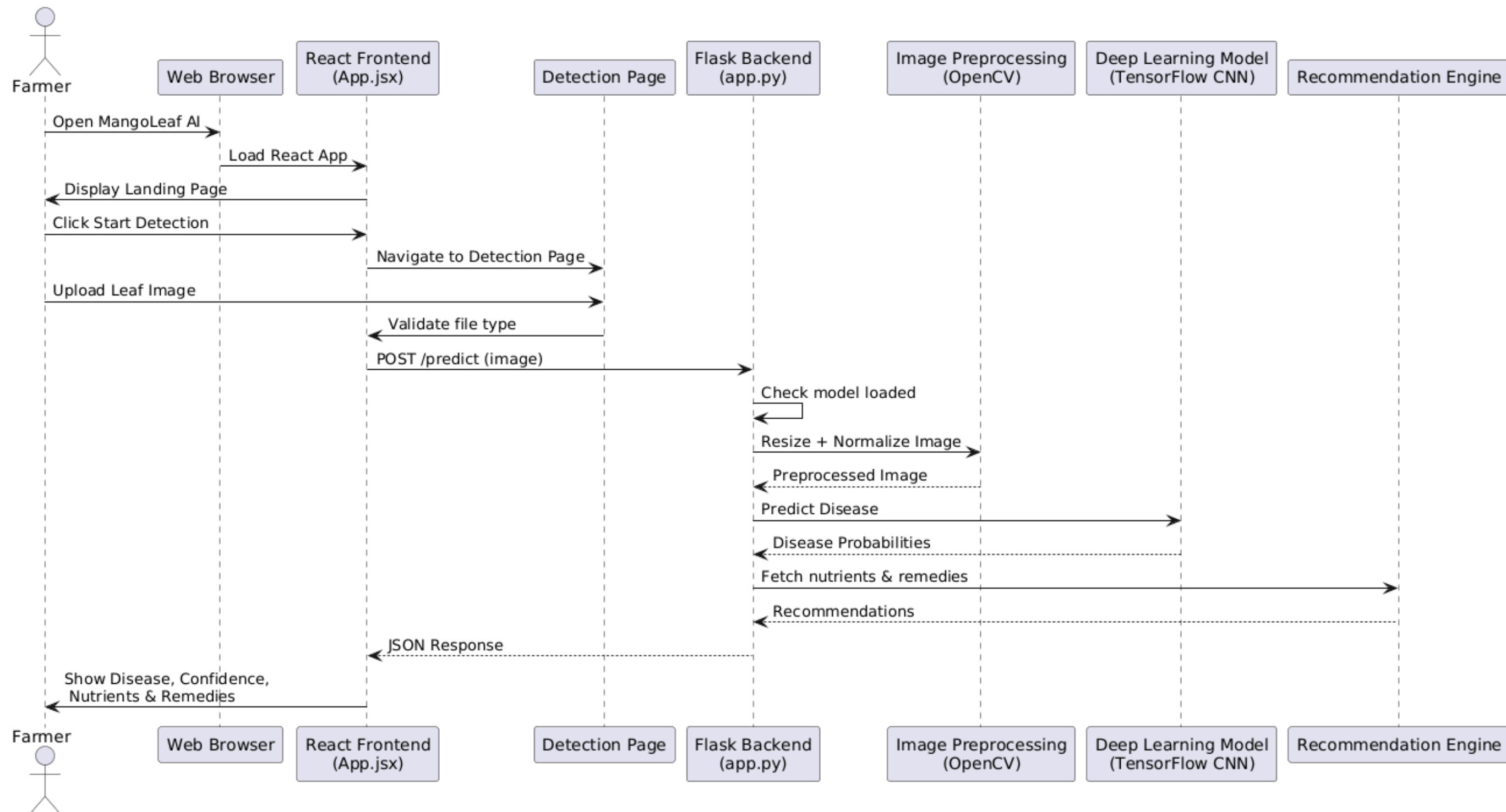
Use Case diagram



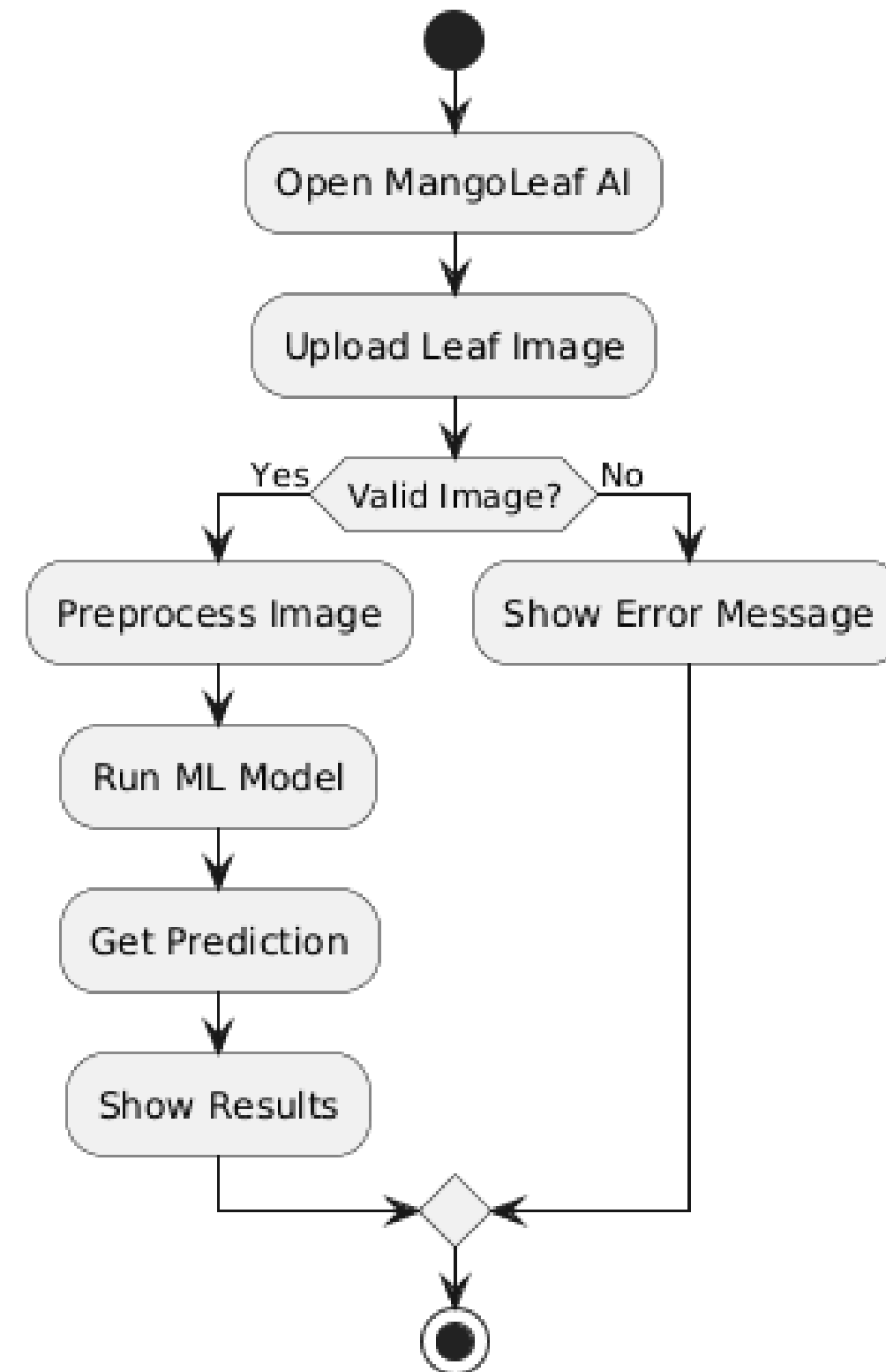
Class diagram



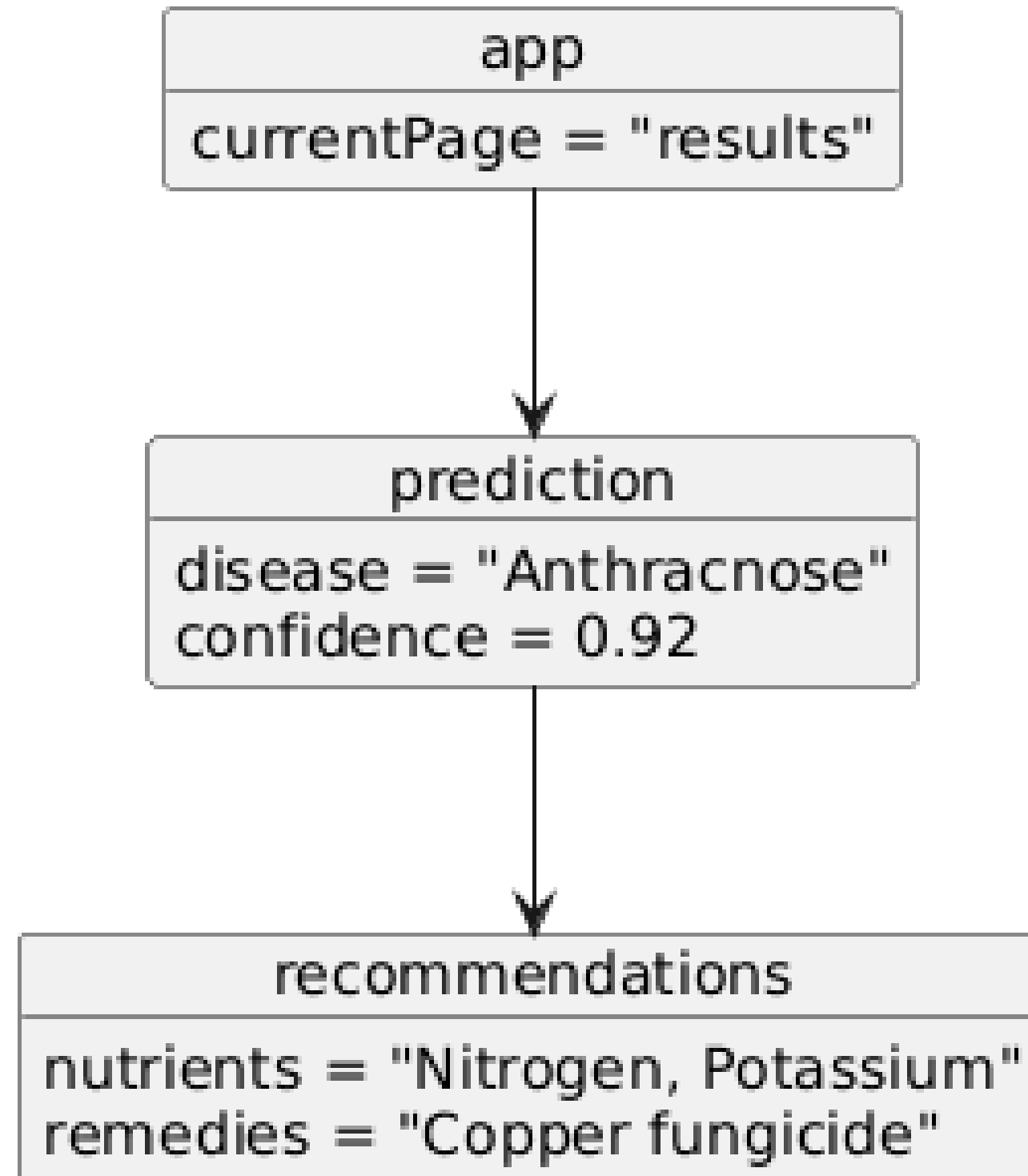
Sequence diagram



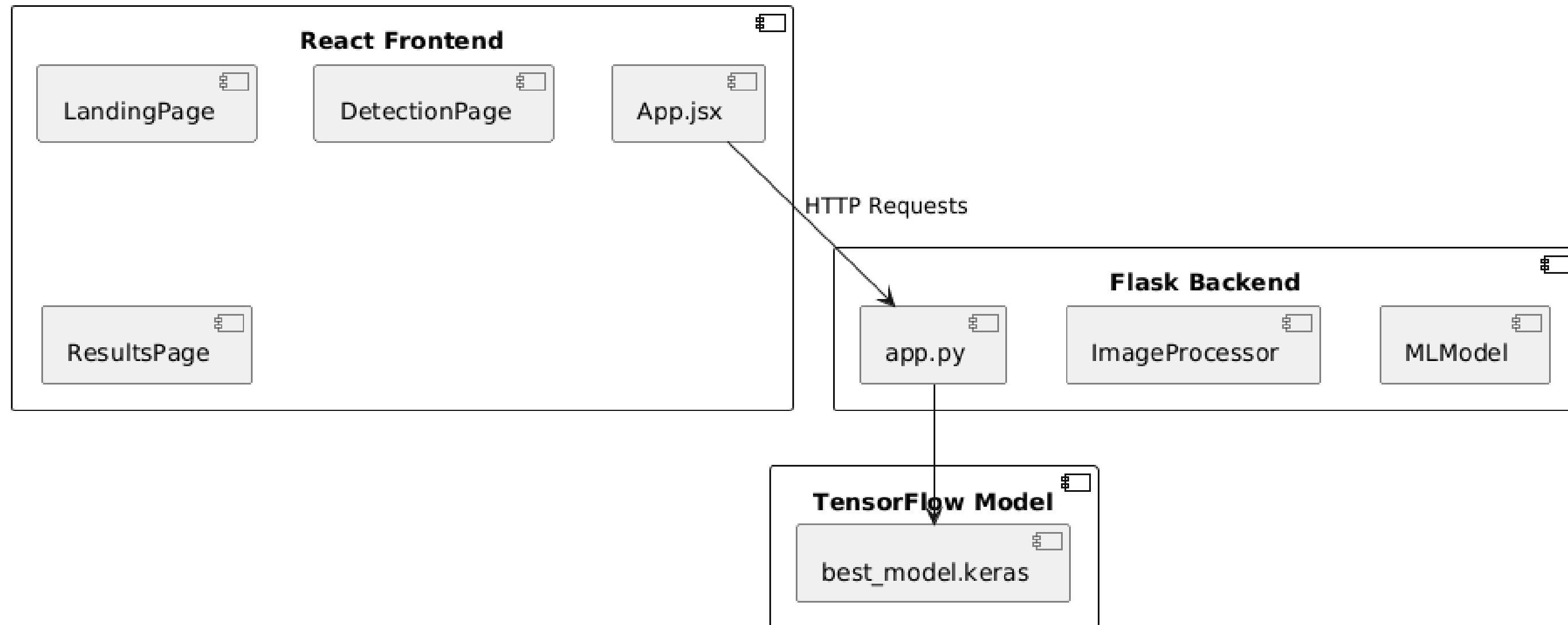
Activity diagram



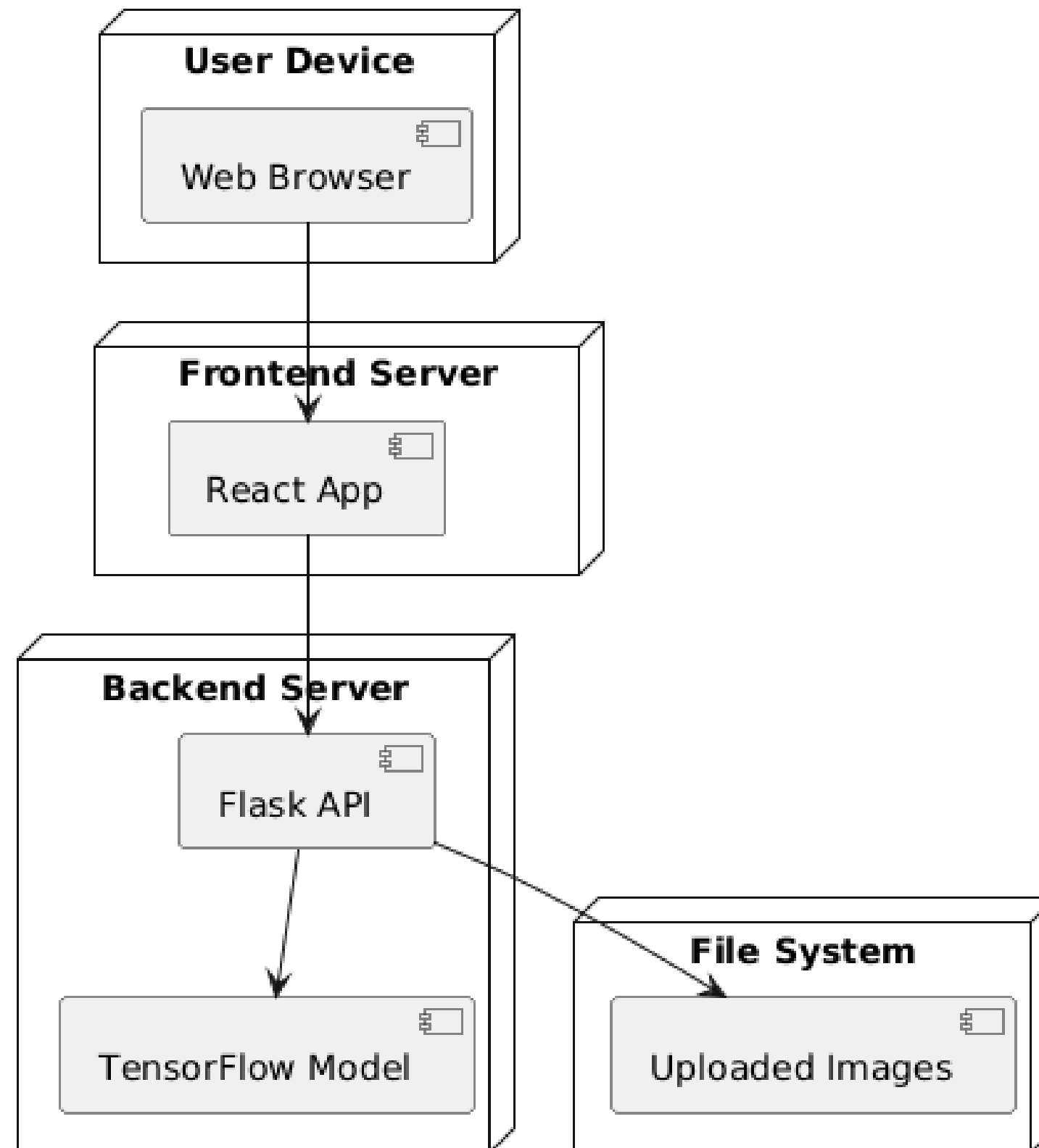
Object diagram



Component diagram



Deployment diagram



Results

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MangoLeaf AI

Protect your mango harvest with AI-powered disease detection. Upload a leaf image and get instant diagnosis with expert recommendations.

[Start Detection](#)

AI Analysis

Advanced deep learning models trained on thousands of mango leaf images for accurate disease identification.



Instant Results

Get immediate diagnosis and treatment recommendations in seconds, not days or weeks.



Expert Guidance

Receive detailed treatment plans and nutrient recommendations based on detected conditions.

Conclusion

- Mango leaf diseases and nutrient deficiencies significantly impact crop yield
- Early and accurate detection is essential for effective crop management
- AI-based image analysis provides a promising solution for automated diagnosis
- The proposed system aims to identify multiple diseases and deficiencies from leaf images
- Mobile-friendly deployment can support farmers with timely decision-making
- This approach has the potential to improve productivity and promote sustainable agriculture



References

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[5] **M. T. Islam et al.**, **MangoLeafDB**: Dataset for mango leaf disease classification, Mendeley Data, 2024.

DOI: <https://doi.org/10.17632/7ghdbftp54.1>

Thank You

